

Update on 2025 and 2026 tracking efficiencies

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Tag-and-probe method

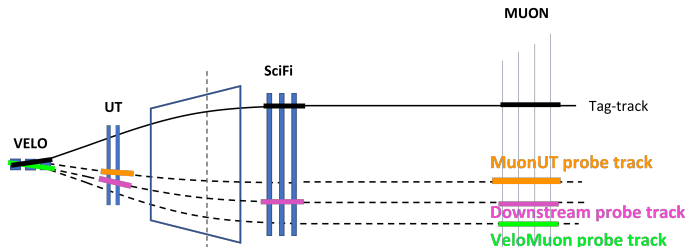


Figure by Rowina

- Tag-and-probe method with $J/\psi \rightarrow \mu^+ \mu^-$
- Fully reconstruct one muon...
- ... and partially reconstruct the other muon
- Match hits in specific sub-detector with partially reconstructed track

$$\epsilon_{\text{track}} = \frac{N_{\text{matched}}}{N_{\text{matched}} + N_{\text{failed}}}$$

Strategy for tracking efficiency determination:

- Combine efficiencies of two methods:
 - VeloMuon: SciFi efficiency
 - Downstream: Velo efficiency
 - Combined: $\text{Velo} \times \text{SciFi}$ efficiencies
- Cross check with long track efficiency from the MuonUT method
 - Main systematic

Overview of work

Recent presentations:

- [A/S week](#)
 - 2024 v0 tables available on GitLab [here](#)
- [RTA WP4 talk on muon tracking efficiencies](#)
- [RTA WP4 talk on hadronic corrections](#)

Tracking efficiencies are determined using TrackCalib2

- GitLab repository [here](#)
- Mass fit done on tuples from AP to extract efficiencies
- Note: Analysts should not have to run TrackCalib2 themselves, unless the released numbers are not suitable (need different binning, etc)

Datasets considered:

• 2024

- v0 results for blocks 2, 1, 5, 6, 7, 8, *pp* ref available
- Currently waiting for large MC production (16M per block) to finish
- Blocks 4 and 3 don't have MC yet

• 2025

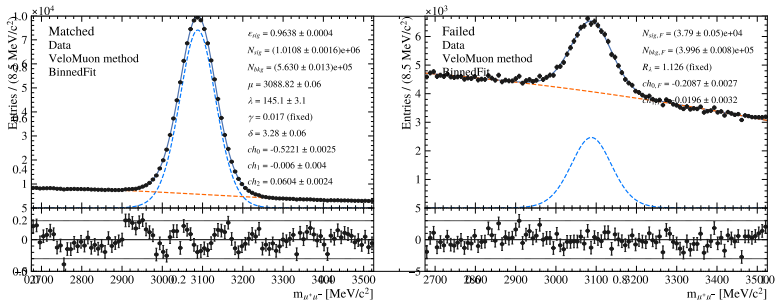
- Today: Analysis of pre-TS data
- Post-TS MC in production (4M per block)
- Very small (100k) post-TS MC samples available for initial checks

• 2026

- Today: First look at data
- Realistic MC not available yet

Example mass fit

Example of fit to extract efficiency (2024 block 5, 1D η bin 3)



- Efficiency is a free parameter in the fit
- Matched and failed samples simultaneously fitted to ensure correct uncertainties

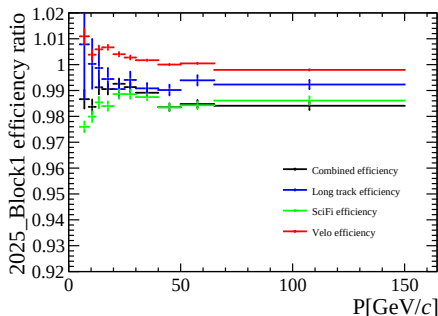
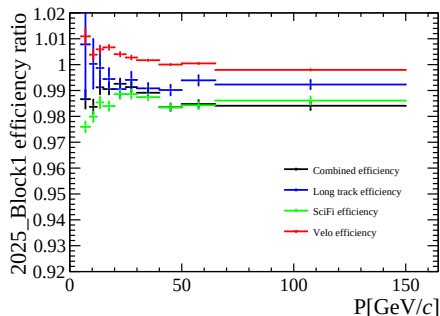
2025 block definitions

Define 2025 blocks according to simulation conditions:

Block	Sprucing campaign	Magnet polarity	MC sample	$\mathcal{L}_{\text{int}}(\text{fb}^{-1})$
1	Sprucing25c1	MagDown	W21	0.4
2	Sprucing25c1	MagUp	W22.25	1.6
3	Sprucing25c2	MagDown	W28.36	4.3
4	Sprucing25c3/4	MagUp	W37.43	4.1
5	Sprucing5c4	MagDown	W43.45	1.0

2025 data/MC ratios in 1D

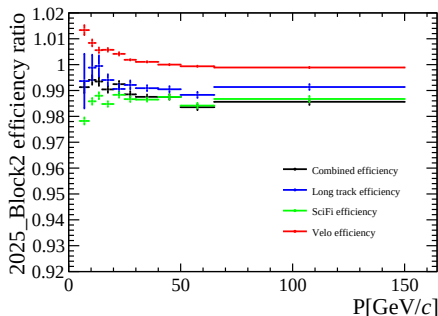
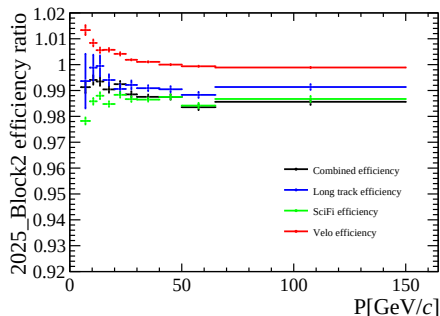
2025 pre-TS data/MC ratios, Block 1



Note that Velo efficiencies are underestimated for this block!

2025 data/MC ratios in 1D

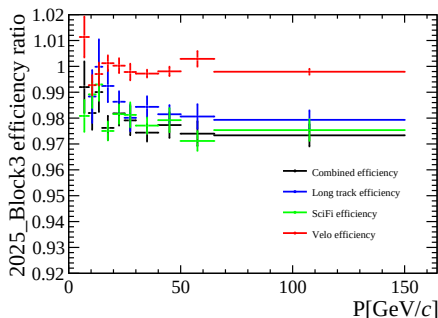
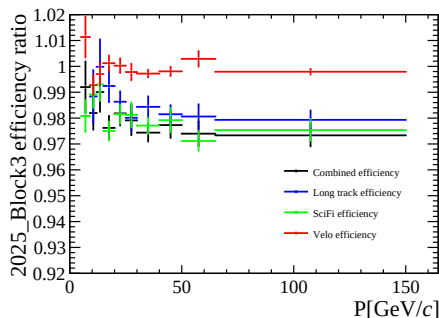
2025 pre-TS data/MC ratios, Block 2



Note that Velo efficiencies are underestimated for this block!

2025 data/MC ratios in 1D

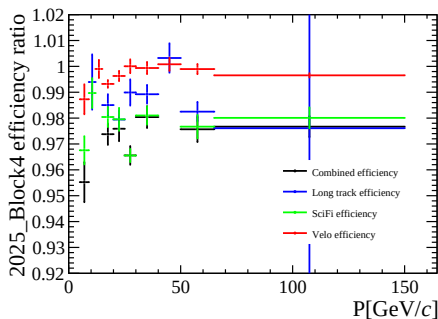
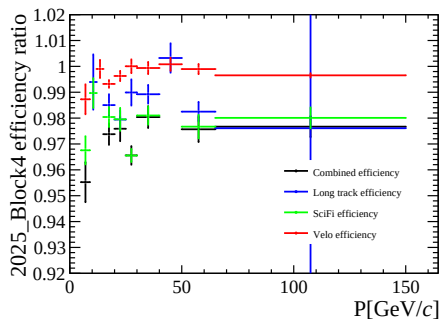
2025 pre-TS data/MC ratios, Block 3



Better Velo efficiency agreement, but MC statistics very low...

2025 data/MC ratios in 1D

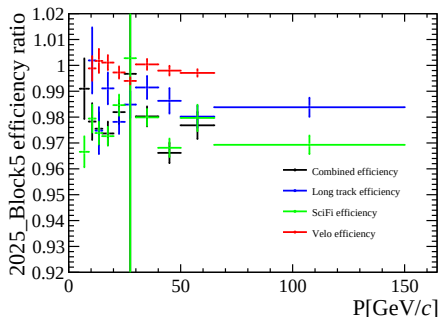
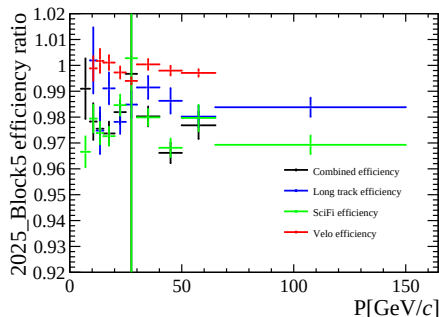
2025 pre-TS data/MC ratios, Block 4



Better Velo efficiency agreement, but MC statistics very low...

2025 data/MC ratios in 1D

2025 pre-TS data/MC ratios, Block 5

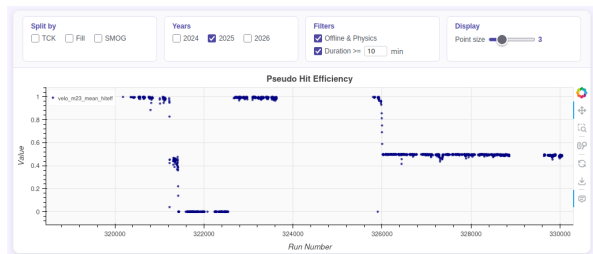


Better Velo efficiency agreement, but MC statistics very low...

2025 pre-TS Velo discrepancies

Why are Velo efficiencies underestimated in pre-TS MC?

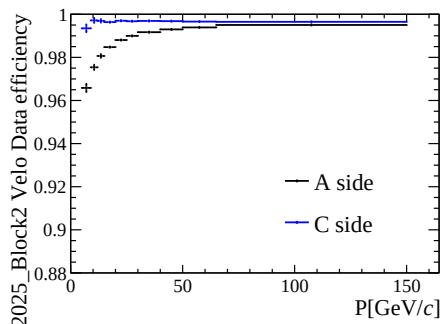
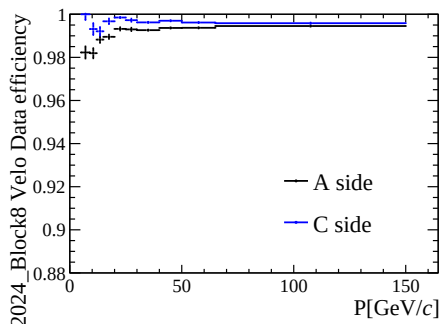
- Velo M23 was (sometimes) working in data, but all four sensors are turned off in MC
- Two sensors in Velo M21 were turned off
- Two consecutive modules that are not working properly can significant impact track efficiency!



M23 hit efficiency during data taking from Intelli RTA

2025 pre-TS Velo discrepancies

Note: Both M21 and M23 are on the A-side, leading to a significant left-right asymmetry



Only M23 was off in 2024, but in 2025 two sensors failed in M21

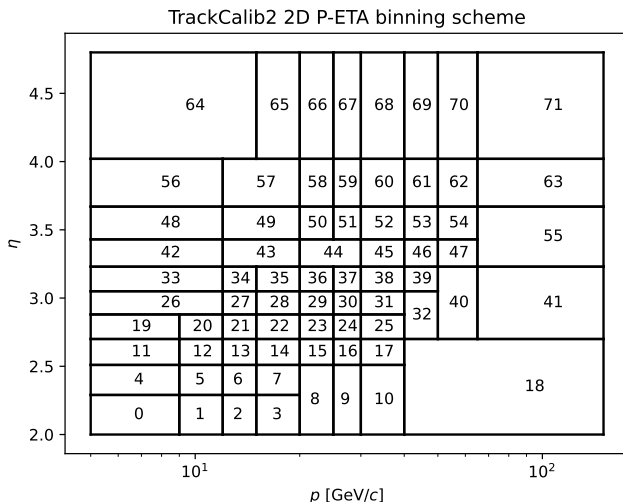
What are the consequences for tracking efficiencies?

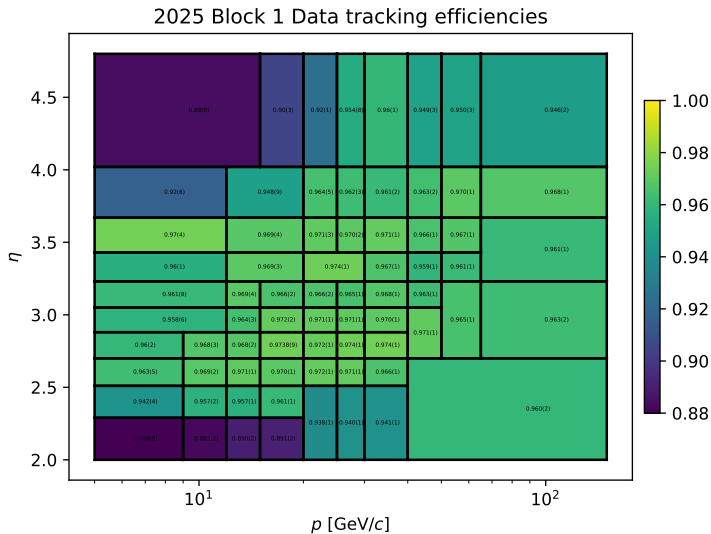
- All Sim10g MC samples are affected
- In principle tracking efficiency corrections should account for this...
- ... but a fix is also in progress: [!306](#)

When the larger MC samples are available, we propose the following mitigation: Change p - η binning to p - η - ϕ with two ϕ bins corresponding to left and right ($|\phi| < \pi/2$ and $|\phi| \geq \pi/2$)

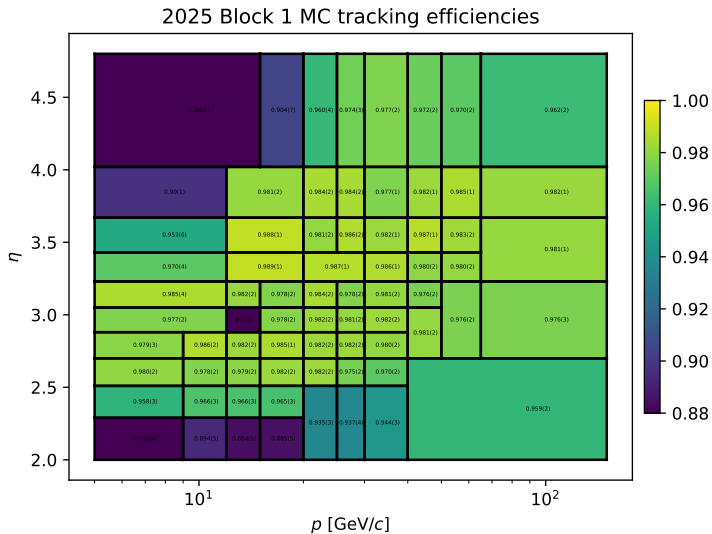
2D binning

2D binning in p vs η with 72 bins:

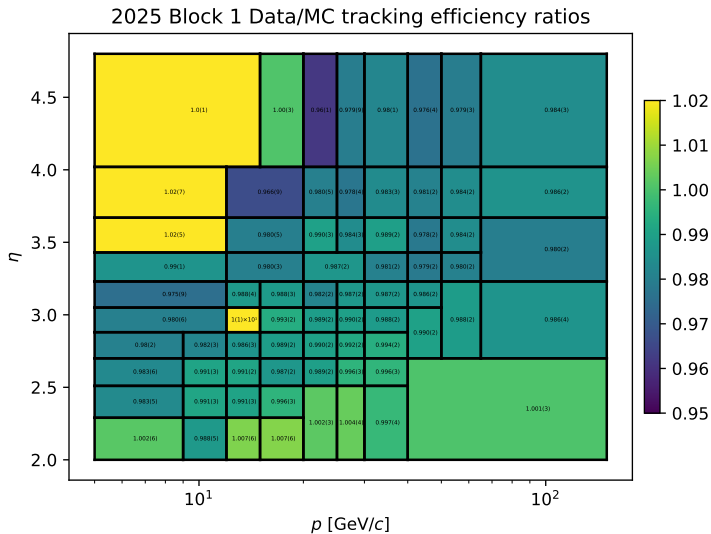




2D binning

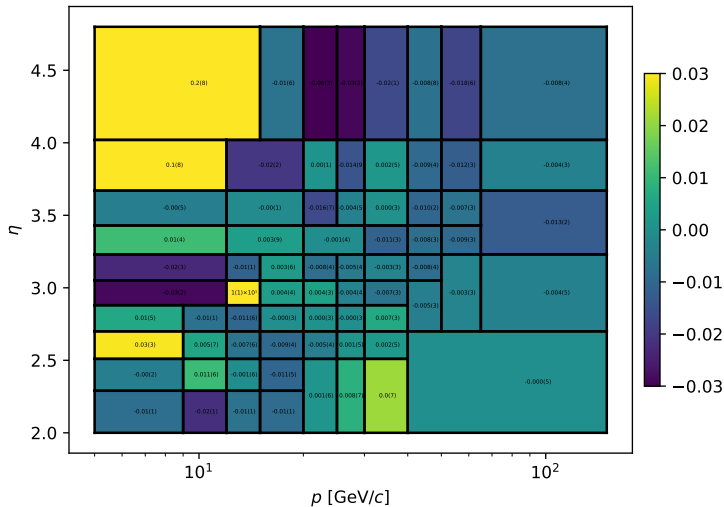


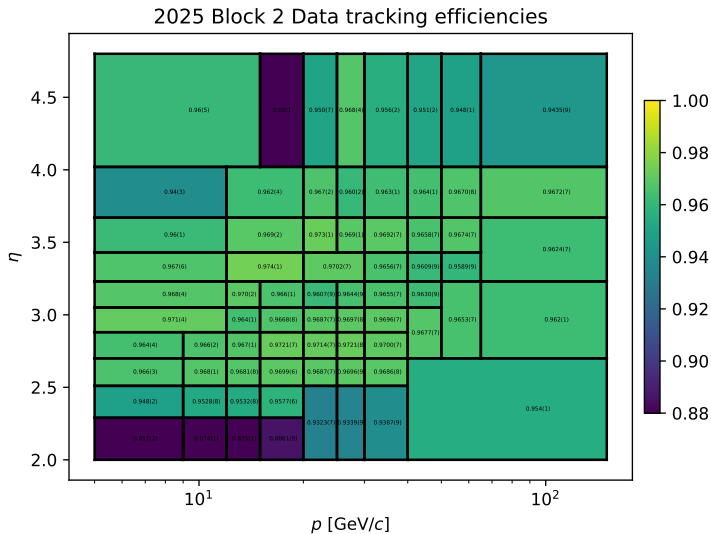
2D binning

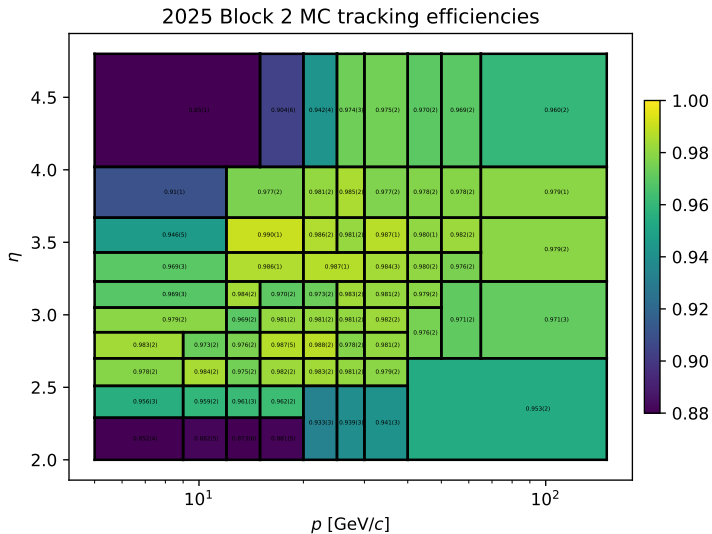


2D binning

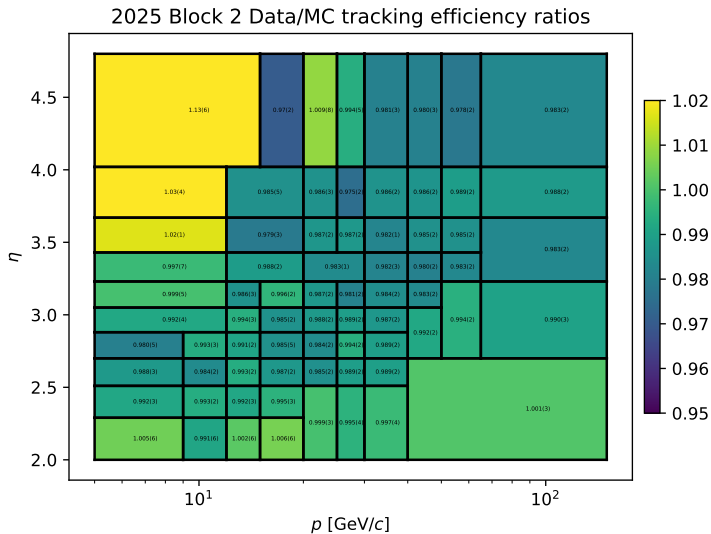
2025 Block 1 Data/MC Combined vs MuonUT ratio difference





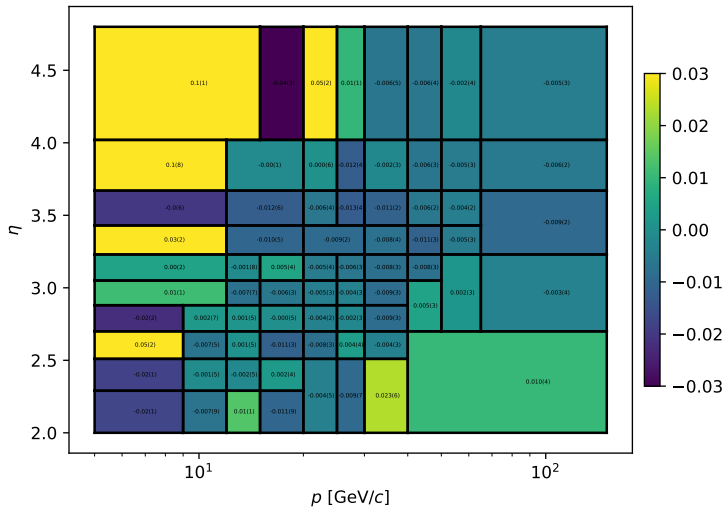


2D binning



2D binning

2025 Block 2 Data/MC Combined vs MuonUT ratio difference

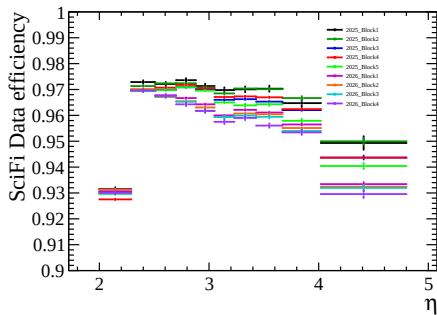
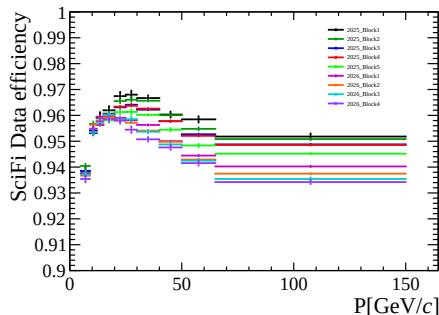


We would like to release the tracking efficiency correction tables for 2025 pre-TS

- 2024 v0 is already available on [GitLab](#)
- Add a file for 2025 with the same binning and same format
- Feedback is welcome!

Initial look at 2026 efficiencies

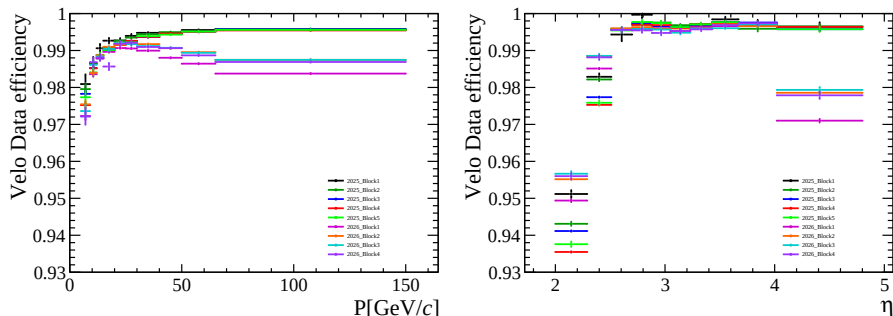
Data SciFi efficiency comparison between 2025 and 2026



- Drop in efficiencies over time, likely due to radiation damage
- Effect greater in region close to the beam (high η)

Initial look at 2026 efficiencies

Data Velo efficiency comparison between 2025 and 2026



- Clear drop in efficiencies in 2026 at high p and in highest η bin
- Most likely due to damage suffered during 2025 heavy ion run

Summary and next steps

Summary:

- ① Behaviour of 2025 tracking efficiencies understood
- ② Correction tables for pre-TS have been prepared
- ③ Post-TS tables will follow once MC is ready
- ④ See clear drop in Velo efficiencies in 2026 due to damage

Next steps

- Release 2025 post-TS tables (and 2026) once MC is available
- Finalise hadronic corrections
- Update tables using larger MC samples
 - Revised P-ETA binning
 - Separate tables for μ^+ and μ^-
 - Charge integrated table with left and right bins in ϕ

Thanks for your attention!